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Review article

The efficacy of colorectal cancer screening: a systematic review and meta-analysis

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ABSTRACT

Objective: To estimate sensitivity and specificity for two major colorectal cancer screening tests in asymptomatic average-risk adults: fecal immunochemical testing and colonoscopy.

Methods: This review was registered in the International Prospective Register of Systematic Reviews (CRD420250635078) on 7 January 2025 and conducted according to Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. PubMed and Scopus were searched without publication date restrictions from database-specific inception dates to 26 June 2025: PubMed from 1946 and Scopus from 1788. Randomized trials and observational studies were screened using predefined eligibility criteria. Risk of bias was assessed using Cochrane tools and the Risk of Bias in Non-randomized Studies of Interventions tool.

Results: Seventeen studies met inclusion criteria. Fecal immunochemical testing had a pooled sensitivity of 77% (95% CI: 67%–85%) and specificity of 95% (95% CI: 93%–96%). Colonoscopy showed higher performance, with sensitivity of 97% (95% CI: 91%–99%) and specificity of 99% (95% CI: 96%–100%). Subgroup analyses indicated variation in fecal immunochemical testing sensitivity across study designs and income settings, whereas specificity remained high.

Conclusions: Colonoscopy remains the benchmark for colorectal cancer detection, while fecal immunochemical testing is a practical, non-invasive option where colonoscopy capacity is limited.

1. Introduction

Colorectal cancer ranks among the three most commonly diagnosed malignancies worldwide and is currently the second leading cause of cancer-related mortality (Sung et al., 2021). In 2020 alone, global estimates reported more than 1.9 million new colorectal cancer cases and nearly 935,000 deaths (Sung et al., 2021). Although the majority of colorectal cancer cases (around 90%) are diagnosed in individuals over the age of 50, recent epidemiological evidence shows a concerning rise in incidence among younger adults aged 20–49 years (Sung et al., 2021). Since the early 1990s, incidence rates in this age group have almost doubled, with annual increases ranging from 1.6% to 7.9% (Vuik et al., 2019; Akimoto et al., 2021; Dharwadkar et al., 2022). This trend has been attributed to lifestyle changes, dietary habits, and genetic predispositions, emphasizing the need for accessible and effective screening strategies to reduce preventable morbidity and mortality.

Screening for colorectal cancer is recognized as one of the most effective public health measures for lowering both disease incidence and mortality. By enabling the early detection and removal of precancerous adenomas, as well as the diagnosis of malignancies at earlier, more treatable stages, screening substantially improves population-level outcomes. Among currently available methods, the fecal immunochemical test and colonoscopy are the most widely implemented worldwide. Fecal immunochemical test, a non-invasive stool-based assay designed to detect occult blood, is particularly well suited for large-scale population screening due to its simplicity, affordability, and adaptability in both primary care and resource-limited contexts (Shaukat and Levin, 2022). However, the fecal immunochemical test demonstrates moderate diagnostic performance, with an estimated sensitivity of about 79%, 95% CI: “69%, 86%” and a specificity of 94%, 95% CI: “92%, 95%” for identifying colorectal cancer (Lee et al., 2014). These values imply a false-negative rate of around 21% and a false-positive rate of about 6%, which may lead to missed diagnoses or unnecessary follow-up colonoscopies in healthy individuals. By contrast, colonoscopy, while invasive, allows direct visualization of the entire colon and rectum and facilitates polyp removal or biopsy during the same procedure. Its diagnostic accuracy is superior, with sensitivity exceeding 95% and specificity approaching 100% (Lin et al., 2021; Rex

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et al., 2017) resulting in a false-negative rate below 5% and a near-zero false-positive rate, making it the gold standard for confirming positive screening tests (Rex et al., 2017).

Implementation of colorectal cancer screening varies considerably across countries, influenced by health system capacity, available resources, and national policies. Fecal immunochemical test is the predominant tool in many European, Asian, and Latin American countries, achieving participation rates exceeding 60% through nationwide organized programs (Shaukat and Levin, 2022). In contrast, countries such as the United States, Germany, and Japan primarily rely on colonoscopy, with uptake ranging from 63 to 67%, 45–61%, and 10–12%, respectively (Shaukat and Levin, 2022; Hotta et al., 2016; Joseph et al., 2020; Starker et al., 2017). Despite its diagnostic advantages, colonoscopy uptake is often constrained by barriers such as bowel preparation requirements, discomfort, fear of the procedure, and time away from work.

Several meta-analyses have previously evaluated the diagnostic accuracy of fecal immunochemical test and colonoscopy for colorectal cancer detection, often reporting high sensitivity estimates for fecal immunochemical test. However, many of these reviews included heterogeneous study designs, such as case–control studies, symptomatic populations, or mixed-risk cohorts, which may overestimate diagnostic performance and limit applicability to population-based screening settings.

Several prior evaluations have reported relatively high sensitivity estimates for the fecal immunochemical test, often ranging between approximately 85% and 92% (Lin et al., 2021; Rex et al., 2017; Tran et al., 2022). However, these estimates were predominantly derived from studies conducted in specific healthcare contexts, such as organized screening programs in high-income countries, selective or highly adherent populations, or settings using optimized fecal immunochemical test thresholds and standardized testing protocols (Lee et al., 2014; Rex et al., 2017; Canévet et al., 2023; Van der Vlugt et al., 2022). Consequently, these results may not fully reflect the diagnostic performance of fecal immunochemical test when implemented across diverse real-world populations and healthcare systems (Van der Vlugt et al., 2022).

The present systematic review and meta-analysis address these limitations by focusing specifically on asymptomatic, average-risk adults undergoing colorectal cancer screening and by synthesizing evidence from contemporary real-world screening studies. By applying stricter inclusion criteria and updated analytical methods, this study provides more conservative and policy-relevant estimates of fecal immunochemical test sensitivity and specificity, thereby clarifying the performance of fecal immunochemical test in routine screening practice and informing screening strategy decisions, particularly in resource-limited settings.

Given these differences, assessing the comparative performance of fecal immunochemical test and colonoscopy—particularly their diagnostic accuracy and feasibility across different healthcare systems—remains an important research priority. To address this, we carried out an updated systematic review and meta-analysis with the objective of integrating global evidence on the sensitivity and specificity of these two screening methods for colorectal cancer.

2. Methods

The study protocol was prospectively registered in PROSPERO (CRD420250635078) on 7th January 2025 to ensure transparency and reproducibility of methods. All stages of the review process were implemented following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) recommendations (Liberati et al., 2009).

2.1. Search strategy for identification of studies

A systematic literature search was conducted in PubMed and Scopus. No restriction on publication date was applied. Each database was searched from its database-specific earliest available coverage date to the final search date of 26 June 2025. Specifically, PubMed was searched from 1946 to 26 June 2025, and Scopus was searched from 1788 to 26 June 2025. All retrieved records were assessed according to the predefined inclusion and exclusion criteria. The search strategy was developed according to the PICO framework:

Population: individuals with colorectal conditions, “colorectal cancer,” asymptomatic adults, or those at “average risk.”

Intervention: screening with fecal immunochemical test or colonoscopy.

Comparator: no screening or control groups.

Outcome: diagnostic accuracy measures, including sensitivity, specificity, and receiver operating characteristic values.

The combined strategy used both controlled vocabulary and free-text terms related to colorectal cancer, risk profiles, fecal immunochemical test, colonoscopy, and diagnostic performance outcomes.

2.2. Study selection

Eligible studies included observational designs, specifically cohort and cross-sectional investigations, that satisfied the following criteria: (Sung et al., 2021) adult participants at average risk or without symptoms of colorectal cancer; (Vuik et al., 2019) direct comparison of fecal immunochemical test and colonoscopy; and (Akimoto et al., 2021) reporting of diagnostic outcomes such as sensitivity, specificity, or overall test accuracy for these modalities. The eligible adult population was defined as individuals aged 40 to 80 years.

2.3. Interventions and outcomes of interests

This review focused on two principal approaches endorsed for large-scale colorectal cancer screening. Fecal immunochemical test is a stool-based method that identifies microscopic quantities of human hemoglobin through antibody-based detection. Owing to its non-invasive nature, affordability, and ease of integration into health programs, fecal immunochemical test is recommended for individuals at average risk, typically administered annually or biennially (Lin et al., 2021; American Cancer Society, 2024). In contrast, colonoscopy is an endoscopic technique that enables direct visualization of the entire colon and rectum using a flexible, camera-equipped device. In addition to detecting neoplastic lesions, colonoscopy offers the advantage of immediate removal of adenomatous polyps and the biopsy of suspicious lesions during the same procedure, thereby combining diagnostic and preventive functions (Shaukat and Levin, 2022; Maida et al., 2024). Together, these two approaches represent the cornerstone of contemporary colorectal cancer screening strategies worldwide.

The main outcomes evaluated were measures of diagnostic accuracy relevant to colorectal cancer screening. Sensitivity was defined as the proportion of individuals with colorectal cancer who were correctly classified by the test, while specificity referred to the proportion of individuals without the disease who were accurately identified (Lee et al., 2014).

2.4. Data collection and risk of bias evaluation

Two reviewers (SL and NP) independently examined titles, abstracts, and full texts to determine eligibility. From each study that met the inclusion criteria, we extracted information on study design, population characteristics (sample size, age, sex distribution), screening method, and diagnostic performance metrics (sensitivity, specificity, true/false positives, and true/false negatives).

The methodological quality of non-randomized studies was evaluated using the ROBINS-I tool. This framework assesses risk of bias across seven domains, including participant selection, confounding, classification of interventions, deviations from intended interventions, missing

data, outcome measurement, and selective reporting. Each domain was rated as having low, moderate, serious, or critical risk of bias.

2.5. Statistical analysis

We conducted a meta-analysis to calculate pooled estimates of diagnostic accuracy for both fecal immunochemical test and colonoscopy. For each screening method, reported numbers of true positives, false positives, true negatives, and false negatives were used to derive sensitivity, specificity, and the diagnostic odds ratio. To account for the inherent correlation between sensitivity and specificity, a bivariate random-effects model was applied.

Statistical heterogeneity across studies was evaluated using Cochrane's Q test and quantified with the I^2 statistic. Depending on the magnitude of heterogeneity, either a fixed-effect or a random-effects model was applied. To investigate potential sources of variation, subgroup analyses were conducted by study design, participant age, and geographic setting. Possible publication bias was assessed through Deek's funnel plot asymmetry test, and a *p*-value below 0.05 was considered evidence of statistical significance.

Statistical analyses were performed using **Stata®** software (version 17; StataCorp, College Station, Texas, United States).

2.6. Ethical considerations

This systematic review and meta-analysis used publicly available, anonymized data; therefore, ethical approval and informed consent were not required.

3. Results

The study selection process is presented in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram (Supplement Fig. 7). In total, 10,127 records were retrieved from the database searches, including 6087 from PubMed and 4040 from Scopus. Following the removal of 3628 duplicates, 6499 unique records were screened. Of these, 6456 articles were excluded as they did not fulfill the PICO-based eligibility criteria. Ultimately, 17 studies (Areia et al., 2019; Aronsson et al., 2017; Barouni et al., 2012; Cenin et al., 2022; Cenin et al., 2020; Fisher et al., 2021; Goede et al., 2017; Heitman et al., 2010; Jahn et al., 2019; Lu et al., 2024; Melnitchouk et al., 2018; Nascimento de Lima et al., 2025; Phisalprapa et al., 2019; Sekiguchi et al., 2016; Tsapournas et al., 2020; Van den Puttelaar et al., 2024; Zauber, 2010) were deemed eligible and included in the final meta-analysis.

3.1. Characteristics of included studies and risk of bias

Seventeen studies (Areia et al., 2019; Aronsson et al., 2017; Barouni et al., 2012; Cenin et al., 2022; Cenin et al., 2020; Fisher et al., 2021; Goede et al., 2017; Heitman et al., 2010; Jahn et al., 2019; Lu et al., 2024; Melnitchouk et al., 2018; Nascimento de Lima et al., 2025; Phisalprapa et al., 2019; Sekiguchi et al., 2016; Tsapournas et al., 2020; Van den Puttelaar et al., 2024; Zauber, 2010) satisfied the eligibility criteria. Among these, five (Heitman et al., 2010; Phisalprapa et al., 2019; Sekiguchi et al., 2016; Tsapournas et al., 2020; Zauber, 2010) used a cross-sectional design, whereas the remaining twelve (Areia et al., 2019; Aronsson et al., 2017; Barouni et al., 2012; Cenin et al., 2022; Cenin et al., 2020; Fisher et al., 2021; Goede et al., 2017; Jahn et al., 2019; Lu et al., 2024; Melnitchouk et al., 2018; Nascimento de Lima et al., 2025; Van den Puttelaar et al., 2024) employed a cohort design. Of the included research, ten studies (Areia et al., 2019; Aronsson et al., 2017; Cenin et al., 2020; Fisher et al., 2021; Goede et al., 2017; Heitman et al., 2010; Jahn et al., 2019; Nascimento de Lima et al., 2025; Sekiguchi et al., 2016; Tsapournas et al., 2020; Van den Puttelaar et al., 2024; Zauber, 2010) were carried out in high-income settings such as the United States, Canada, Japan, Australia, Austria, Portugal, Sweden, and Germany. Conversely, five studies (Barouni et al., 2012; Cenin et al., 2022; Lu et al., 2024; Melnitchouk et al., 2018; Phisalprapa et al., 2019) originated from middle-income regions, including Iran, China, Ukraine, and Thailand.

Across these studies, most participants were between 40 and 80 years of age, consistent with the standard screening age for individuals at average risk. The proportion of male participants varied substantially, ranging from 35.9% to 96.8% (see Table 1).

Using the ROBINS-I tool, the quality of the included studies was appraised. Among the 17 studies, four were judged to have a low risk of bias, eleven were considered at moderate risk, and two were assessed as having a serious risk of bias.

Evidence of publication bias was identified for fecal immunochemical test, as demonstrated by Deek's funnel plot asymmetry test (coefficient = 90.280; sensitivity = 37.936; $p = 0.03$; Supplement fig. 1a). In contrast, no significant evidence of publication bias was found for colonoscopy (coefficient = −397.828; sensitivity = 256.955; $p = 0.15$; Supplement fig. 1b).

3.2. Pooled diagnostic accuracy of fecal immunochemical test and colonoscopy

This meta-analysis incorporated data from 17 eligible studies (Areia et al., 2019; Aronsson et al., 2017; Barouni et al., 2012; Cenin et al., 2022; Cenin et al., 2020; Fisher et al., 2021; Goede et al., 2017; Heitman et al., 2010; Jahn et al., 2019; Lu et al., 2024; Melnitchouk et al., 2018; Nascimento de Lima et al., 2025; Phisalprapa et al., 2019; Sekiguchi et al., 2016; Tsapournas et al., 2020; Van den Puttelaar et al., 2024; Zauber, 2010). The pooled sensitivity for colonoscopy was 97.3%, 95% CI: “91.3%, 99.2%”, reaffirming its consistently high diagnostic accuracy for colorectal cancer (see Table 2, Figs. 3 and 4). By comparison, fecal immunochemical test demonstrated a pooled sensitivity of 76.9%, 95% CI: “66.6%, 84.7%”, supporting its role as a non-invasive test with reasonable case-finding performance (see Table 2, Figs. 1 and 2).

The pooled specificity of colonoscopy was 99.9%, 95% CI: “96.2%, 100%”, confirming its precision in correctly ruling out colorectal cancer (see Table 2; Figs. 3 and 4). The fecal immunochemical test also demonstrated high specificity at 94.7%, 95% CI: “92.9%, 95.9%”, indicating a low rate of false-positive results, (see Table 2, Figs. 1 and 2).

The pooled diagnostic odds ratio for colonoscopy was 54,277.4, 95% CI: “489.7, 601,631.5”, underscoring its strong discriminative ability (see Table 2, Figs. 3 and 4). In comparison, the pooled diagnostic odds ratio for fecal immunochemical test was 58.9, 95% CI: “35.5, 97.8”, indicating moderate diagnostic effectiveness for population-level screening (see Table 2, Figs. 1 and 2).

The pooled positive likelihood ratio for colonoscopy was 1451.9, 95% CI: “24.27, 86,844.50”, reflecting a very strong increase in the probability of colorectal cancer following a positive test result (see Table 2, Figs. 3 and 4). By contrast, the pooled positive likelihood ratio for fecal immunochemical test was 14.4, 95% CI: “35.51, 97.78”, indicating only a moderate rise in disease likelihood after a positive result (see Table 2, Figs. 1 and 2).

For colonoscopy, the pooled negative likelihood ratio was 0.03, 95% CI: “0.01, 0.09”, signifying a minimal chance of missed cases when the result is negative (see Table 2, Figs. 3 and 4). In comparison, fecal immunochemical test yielded a pooled negative likelihood ratio of 0.24, 95% CI: “0.17, 0.36”, suggesting a moderate ability to exclude disease when the test is negative (see Table 2, Figs. 1 and 2).

The summary receiver operating characteristic analysis demonstrated high overall performance for both screening methods. The area under the curve for colonoscopy was 0.99, 95% CI: “0.98, 1.00”, while the receiver operating characteristic for fecal immunochemical test was 0.96, 95% CI: “0.94, 0.97”. These findings confirm that both colonoscopy and fecal immunochemical test are highly accurate procedures for

Table 1
Distribution of study characteristics and diagnostic accuracy estimates among diagnostic accuracy studies of fecal immunochemical testing and colonoscopy for colorectal cancer screening among asymptomatic average-risk adults aged 40–80 years identified from PubMed and Scopus, 4 November 2024 to 26 February 2025.

Author (year)	Study design	Setting	Fecal Immunochemical Test		Colonoscopy		Sample size	Overall mean age	% Male
			Sensitivity	specificity	Sensitivity	specificity			
(Areia et al., 2019)	Cohort	Portugal	0.67	0.98	0.95	NR	NR	60	50
(Aronsson et al., 2017)	Cohort	Sweden	0.88	0.93	0.97	1	1000	60	50
(Barouni et al., 2012)	Cohort	Iran	0.239	0.938	NR	NR	2885	63	96.8
(Cenin et al., 2022)	Cohort	China	0.65	0.929	0.95	0.86	1000	NR	NR
(Cenin et al., 2020)	Cohort	Australia	0.728	0.95	0.95	0.86	800,000	NR	NR
(Fisher et al., 2021)	Cohort	USA	0.738	0.964	0.95	0.86	465	57	45.3
(Goede et al., 2017)	Cohort	Canada	0.83	0.96	0.95	0.9	3790	NR	59.2
(Heitman et al., 2010)	Cross-sectional	USA	0.966	0.95	0.66	1	100,000	NR	40.5
(Jahn et al., 2019)	Cohort	Austria	0.872	0.928	0.947	1	NR	NR	NR
(Lu et al., 2024)	Cohort	China	0.785	0.95	1	1	NR	NR	NR
(Melnitchouk et al., 2018)	Cohort	Ukraine	NR	NR	0.91	0.9	50,148	55.2	35.92
(Nascimento de Lima et al., 2025)	Cohort	USA	0.74	0.96	NR	NR	NR	NR	NR
(Phisalprapa et al., 2019)	Cross-sectional	Thailand	0.67	0.96	0.947	0.998	NR	NR	NR
(Sekiguchi et al., 2016)	Cross-sectional	Japan	NR	0.946	0.999	1	100,000	NR	NR
(Tsapournas et al., 2020)	Cross-sectional	Sweden	0.923	0.773	0.74	NR	355	NR	42.1
(Van den Puttelaar et al., 2024)	Cohort	USA	0.8	0.964	0.91	0.87	NR	NR	59.6
(Zauber, 2010)	Cross-sectional	USA	0.75	0.9	0.95	0.9	NR	NR	NR

NR; No report.

Table 2
Pooled diagnostic accuracy estimates for fecal immunochemical testing and colonoscopy for colorectal cancer detection among asymptomatic average-risk adults aged 40–80 years in diagnostic accuracy studies identified from PubMed and Scopus, 4 November 2024 to 26 February 2025.

Outcome	Coefficient	Standard error	95% CI
Sensitivity			
Fecal immunochemical test	0.77	0.05	0.67, 0.85
Colonoscopy	0.97	0.02	0.91, 0.99
Specificity			
Fecal immunochemical test	0.95	0.01	0.93, 0.96
Colonoscopy	0.99	0.00	0.96, 1.00
Diagnostic Odds Ratio			
Fecal immunochemical test	58.92	15.22	35.51, 97.78
Colonoscopy	54,277.40	130,382.30	489.67, 6016315.00
Positive Likelihood Ratio			
Fecal immunochemical test	14.40	1.97	11.01, 18.84
Colonoscopy	1451.90	3030.71	24.27, 86844.50
Negative Likelihood Ratio			
Fecal immunochemical test	0.24	0.05	0.17, 0.36
Colonoscopy	0.03	0.02	0.01, 0.09

colorectal cancer screening and diagnosis (Figs. 1, 2, 3, and 4).

3.3. Subgroup analyses

In cohort studies, fecal immunochemical test demonstrated a pooled sensitivity of 0.72, 95% CI: "0.61, 0.81" and a pooled specificity of 0.95, 95% CI: "0.94, 0.96" statistically significant with ($p < 0.001$) (Supplement fig. 2). In cross-sectional studies, pooled sensitivity was higher at 0.87, 95% CI: "0.70, 0.95", while pooled specificity was 0.92, 95% CI: "0.84, 0.96", statistically significant with ($p < 0.001$) (Supplement fig. 3).

In high-income countries, fecal immunochemical test achieved a pooled sensitivity of 0.83, 95% CI: "0.76, 0.89" and a pooled specificity of 0.94, 95% CI: "0.92, 0.96" statistically significant with ($p < 0.001$) (Supplement fig. 4). In middle-income countries, fecal immunochemical test sensitivity was notably lower, with a pooled estimate of 0.53, 95% CI: "0.35, 0.70", while specificity remained high at 0.95, 95% CI: "0.94, 0.96" statistically significant with ($p < 0.001$) (Supplement fig. 5). Suggesting considerable variability in diagnostic performance of fecal immunochemical test may affect from the study design and country income level.

In high-income countries, colonoscopy demonstrated consistently high diagnostic accuracy. The pooled sensitivity was 0.96, 95% CI: "0.89, 0.98", and pooled specificity reached 1.00, 95% CI: "0.91, 1.00" statistically significant with ($p < 0.001$) (Supplement fig. 6) reflecting differences in study design, population characteristics, and reference standards. Other subgroup analyses were not provided because of the number of studies included.

4. Discussion

This meta-analysis pooled evidence on the diagnostic performance of fecal immunochemical test and colonoscopy for colorectal cancer screening. Colonoscopy showed consistently high accuracy (sensitivity ~97%, specificity ~100%, receiver operating characteristic 0.99), supporting its role as the reference standard. Fecal immunochemical test demonstrated lower sensitivity (~77%) but maintained high specificity (~95%) and overall good performance (receiver operating characteristic 0.96). Despite its limitations, fecal immunochemical test remains a practical, non-invasive option for population-level screening, particularly where colonoscopy access or adherence is limited.

Our findings are consistent with previous evidence, with fecal immunochemical test sensitivity typically reported between 73% and 88%, aligning with our pooled estimate of 77%. Similar results from large asymptomatic cohorts support the reliability of fecal immunochemical test and reinforce its role in population-based screening (Lee et al., 2014; Lin et al., 2021).

For colonoscopy, our pooled sensitivity of >97% is in line with earlier studies that consistently demonstrate its high diagnostic yield, particularly for detecting (Lee et al., 2014) advanced neoplasia (Maida et al., 2024). The superior sensitivity of colonoscopy is attributable to its ability to directly visualize the mucosa and remove lesions during the same procedure, thereby minimizing the likelihood of missed cases.

Similarly, the pooled specificity of fecal immunochemical test (~95%) is consistent with previous meta-analyses reporting values between 93% and 94% (Lee et al., 2014; Lin et al., 2021). Similarly, colonoscopy showed near-perfect specificity, reinforcing its role as the confirmatory reference standard. This high accuracy likely reflects its capacity for direct mucosal visualization, thereby minimizing false-positive findings.

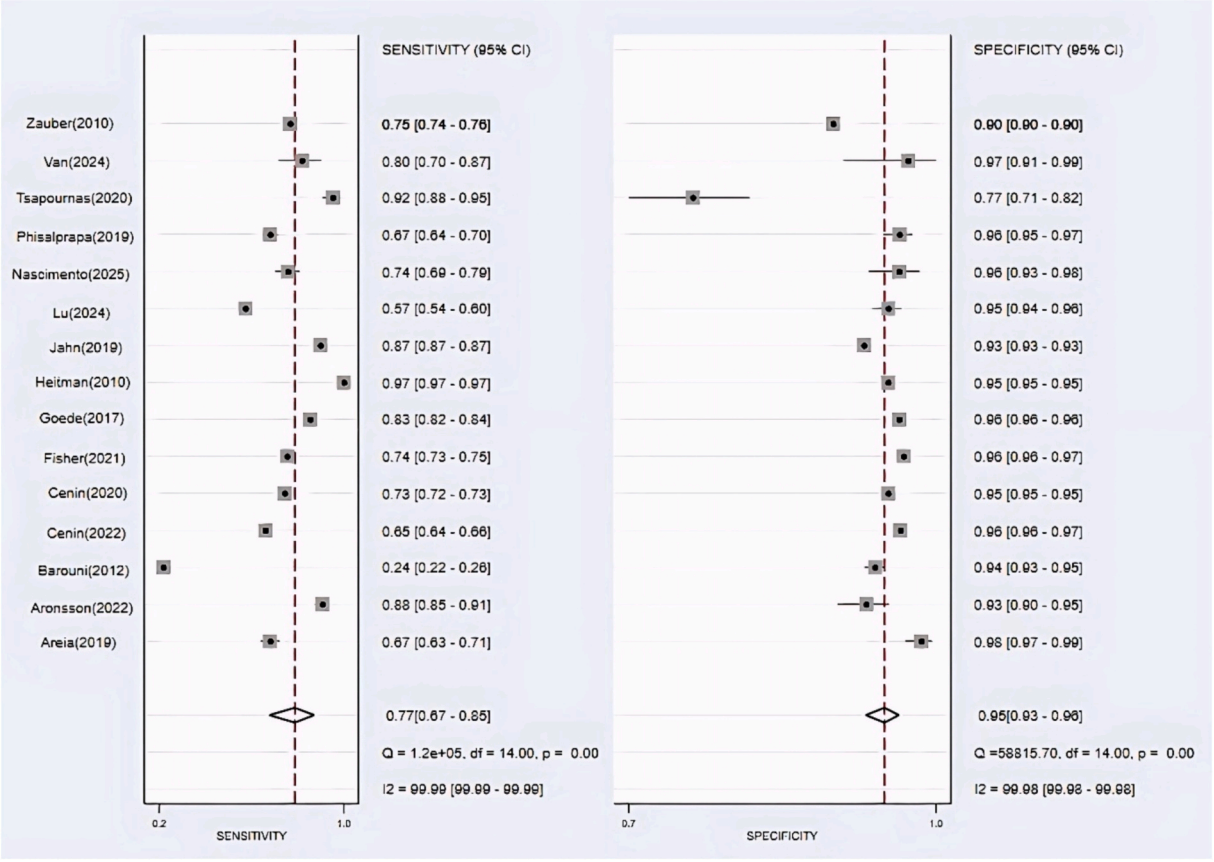


Fig. 1. Forest plots of study-specific sensitivity and specificity with 95% confidence intervals for fecal immunochemical testing to detect colorectal cancer among asymptomatic average-risk adults aged 40–80 years in diagnostic accuracy studies identified from PubMed and Scopus, 4 November 2024 to 26 February 2025.

Additional measures support this pattern. Colonoscopy demonstrated an exceptionally high diagnostic odds ratio (54,277.4), indicating strong discriminative ability, whereas fecal immunochemical test showed a substantially lower diagnostic odds ratio (58.9), reflecting its more moderate performance. As diagnostic odds ratio depends on both sensitivity and specificity—and may vary with prevalence and case definitions—these differences should be interpreted with caution.

Our findings are consistent with previous studies (Areia et al., 2019; Heitman et al., 2010). Reported fecal immunochemical test sensitivity (≈75–81%) and specificity (≈89–95%) align with our pooled estimates (76.9% and 94.7%), and are comparable to USPSTF results (sensitivity 79%, specificity 94%) (Canévet et al., 2023). Similarly, colonoscopy showed consistently high accuracy (sensitivity 97.3%, specificity 99.9%), in line with prior evidence reporting sensitivity >95% and near-perfect specificity (Lee et al., 2014). Overall, these findings support the robustness of our estimates within the range of existing international evidence.

The substantially higher diagnostic odds ratio of colonoscopy likely reflects its ability to directly visualize and remove lesions, reducing both false-positive and false-negative results. Consistently, colonoscopy showed a markedly higher positive likelihood ratio and near-zero negative likelihood ratio, indicating strong rule-in and rule-out performance. In contrast, fecal immunochemical test demonstrated weaker performance, with a higher negative likelihood ratio, partly due to reduced sensitivity for right-sided lesions (Lee et al., 2014).

Previous studies similarly show that colonoscopy achieves very high positive likelihood ratios and low negative likelihood ratios, supporting its strong rule-in and rule-out performance. In contrast, fecal immunochemical test demonstrates more modest positive likelihood ratios and higher negative likelihood ratios, largely reflecting reduced sensitivity for non-bleeding or right-sided lesions (Lee et al., 2014).

Although likelihood ratios, whether positive or negative, are not influenced by disease prevalence and can therefore be generalized across settings, the definition of the target condition must remain consistent with that of the original studies (Lin et al., 2021).

Receiver operating characteristic analysis indicated excellent discrimination for both tests, with colonoscopy performing slightly better. This likely reflects its direct visualization of the colorectal mucosa, allowing detection regardless of bleeding or lesion location. In contrast, fecal immunochemical test relies on occult blood detection, which may reduce sensitivity, particularly for right-sided lesions (Lee et al., 2014).

Our findings align with previous evidence showing that colonoscopy provides superior diagnostic accuracy, while fecal immunochemical test offers a practical balance between sensitivity and ease of implementation (Lee et al., 2014). However, receiver operating characteristic measures reflect overall test performance and do not directly inform rule-in or rule-out decisions; therefore, likelihood ratios should also be considered in clinical interpretation (Lin et al., 2021).

This review has several limitations. First, only observational studies were included, which may limit internal validity. Second, variation in fecal immunochemical test protocols—such as differing thresholds and sampling strategies—may reduce comparability across studies. Third, the predominance of studies from high-income settings may limit generalizability to low- and middle-income contexts. Finally, although the search was limited to January 2025 per protocol, the impact of more recent studies is likely minimal given the stability of diagnostic accuracy estimates.

Moreover, variation in cancer stage may have influenced test performance. Early-stage lesions—particularly flat or non-bleeding tumors located in the right colon—are more often undetected by fecal immunochemical test, since the assay relies on the presence of measurable

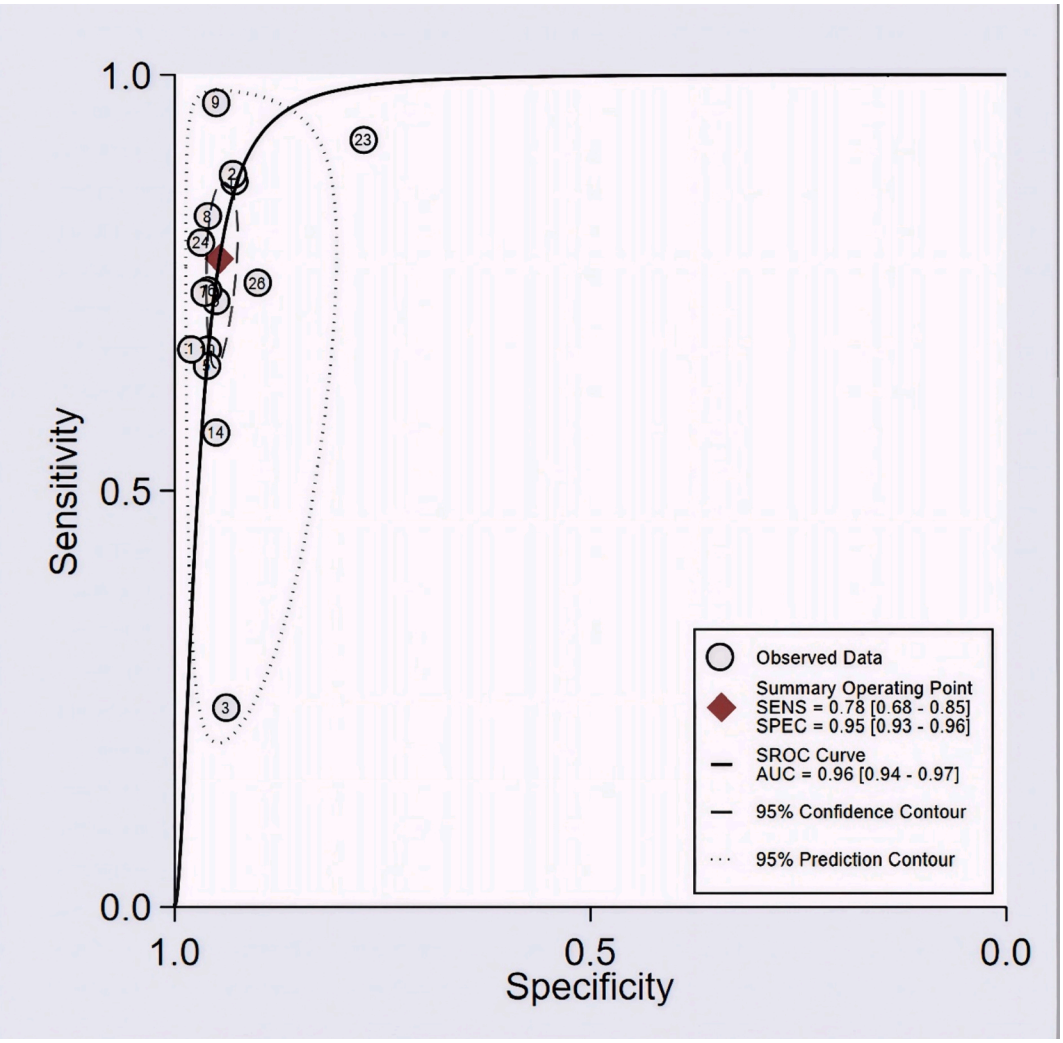


Fig. 2. Summary receiver operating characteristic curve showing the diagnostic performance of fecal immunochemical testing for colorectal cancer detection among asymptomatic average-risk adults aged 40–80 years in diagnostic accuracy studies identified from PubMed and Scopus, 4 November 2024 to 26 February 2025.

hemoglobin in stool (Tran et al., 2022; Van der Vlugt et al., 2022). As staging information was not uniformly reported across all studies, pooled sensitivity estimates may not fully reflect this variability. Colonoscopy accuracy may also have been influenced by factors such as bowel preparation, adenoma detection rates, and operator experience, which likely varied across studies.

Despite these limitations, this meta-analysis has notable strengths. It synthesizes recent real-world evidence comparing fecal immunochemical test and colonoscopy across diverse populations, enhancing external validity. The findings confirm the performance of both modalities and provide a comprehensive assessment using multiple diagnostic indicators (sensitivity, specificity, diagnostic odds ratio, likelihood ratios, and receiver operating characteristic), offering an evidence-based framework for clinical decision-making.

These findings support several practical implications. Fecal immunochemical test should be used as a repeated screening strategy to improve detection over time. Strengthening follow-up systems is essential to ensure completion of diagnostic evaluation after positive results. Screening strategies should be adapted to local contexts, particularly where colonoscopy capacity is limited, with fecal immunochemical test serving as an effective population-level approach. Future policies should also incorporate cost-effectiveness considerations to optimize resource allocation.

Across subgroup analyses, fecal immunochemical test maintained consistently high specificity (>90%), but sensitivity varied, particularly in middle-income and cohort-based studies. In contrast, colonoscopy demonstrated more stable and superior accuracy, reinforcing its role as the diagnostic gold standard.

Substantial heterogeneity was evident across studies. Sample sizes ranged from 355 to approximately 800,000 participants, with seven studies not reporting sample size. Such variation, together with methodological differences, may have influenced the pooled estimates. Although a random-effects model with inverse-variance weighting was applied, the wide disparity in study size—and missing data in several studies—may have introduced bias, with large cohorts disproportionately influencing the results.

Definitions of ‘average risk’ also differed between studies and across health systems. U.S. screening guidelines describe average-risk individuals as adults aged 45 years or older who have no prior colorectal cancer, adenomatous polyps, inflammatory bowel disease, or known hereditary syndromes such as Lynch syndrome. The American Society for Gastrointestinal Endoscopy defines average-risk persons as those aged 45 years and older without previous colorectal cancer or polyps and without any factors that would place them in a high-risk screening category. By contrast, the World Gastroenterology Organization considers men and women with no family history of colorectal cancer to be at average risk and recommends starting colonoscopy at 50 years of age. These differing age thresholds and inclusion/exclusion criteria mean

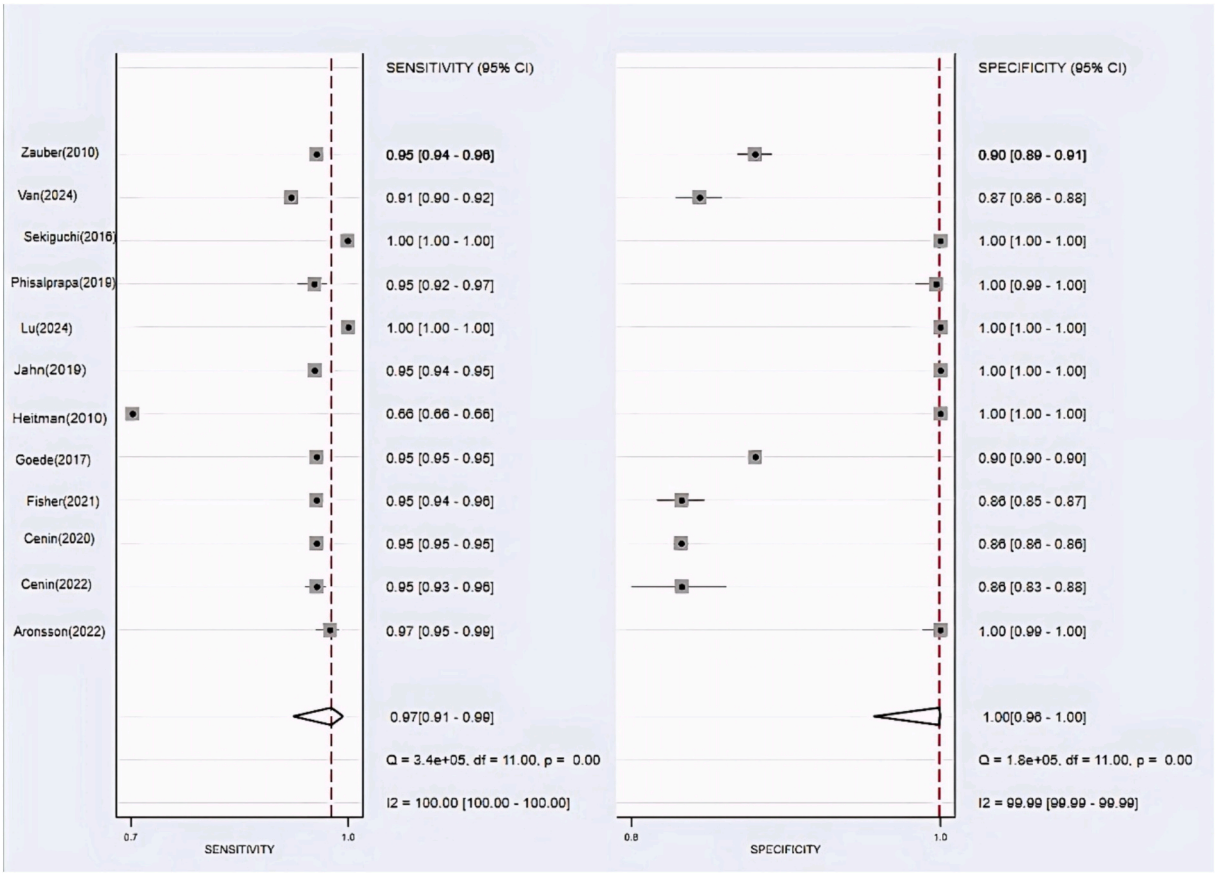


Fig. 3. Forest plots of study-specific sensitivity and specificity with 95% confidence intervals for colonoscopy to detect colorectal cancer among asymptomatic average-risk adults aged 40–80 years in diagnostic accuracy studies identified from PubMed and Scopus, 4 November 2024 to 26 February 2025.

that populations labelled as 'average risk' are not directly comparable. Some studies may have included individuals as young as 45 years or those with a family history, whereas others excluded them. This variability likely contributed to the substantial between-study heterogeneity observed in our meta-analysis.

These findings highlight the need for future meta-analyses to better address heterogeneity through subgroup analyses by study design, sample size, and risk definitions. Improved reporting of sample size and clear criteria for classifying "average risk" are essential. Standardized eligibility definitions in prospective studies would enhance comparability and improve the reliability of pooled estimates.

Unlike earlier meta-analyses that frequently included symptomatic populations or case–control designs, this study synthesizes evidence exclusively from asymptomatic, average-risk screening populations, providing more realistic estimates of fecal immunochemical test performance in real-world screening contexts. The substantially lower pooled sensitivity observed in this analysis compared with prior reviews highlights the impact of study design and population selection on diagnostic accuracy estimates. These findings clarify the discrepancy between optimistic fecal immunochemical test sensitivity reported in earlier literature and the more conservative performance observed in routine screening practice.

Finally, the substantially lower pooled sensitivity observed in this analysis compared with earlier meta-analyses underscores the importance of population selection and study design in estimating real-world screening performance.

5. Conclusions

The findings of this review indicate that colonoscopy continues to serve as the gold standard for colorectal cancer detection, offering consistently high sensitivity and specificity. In contrast, the fecal immunochemical test represents a practical, non-invasive option, with moderate diagnostic accuracy and substantially lower cost, making it particularly appropriate for population-wide screening initiatives.

Maximizing the benefits of both approaches requires attention not only to test accuracy but also to patient adherence, program delivery, and timely follow-up. Integrating these factors into screening strategies is essential to achieve optimal cancer outcomes and ensure that both colonoscopy and fecal immunochemical test contribute effectively to reducing the global burden of colorectal cancer.

CRediT authorship contribution statement

Sai Nyan Lin Tun: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Data curation, Conceptualization. **Noppcha Singweratham:** Validation, Supervision, Methodology, Conceptualization. **Pallop Siewchaisakul:** Project administration, Conceptualization. **Napaphat Poprom:** Writing – review & editing, Visualization, Supervision, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization.

Consent for publication

Not applicable.

Ethics approval and consent to participate

Not applicable.

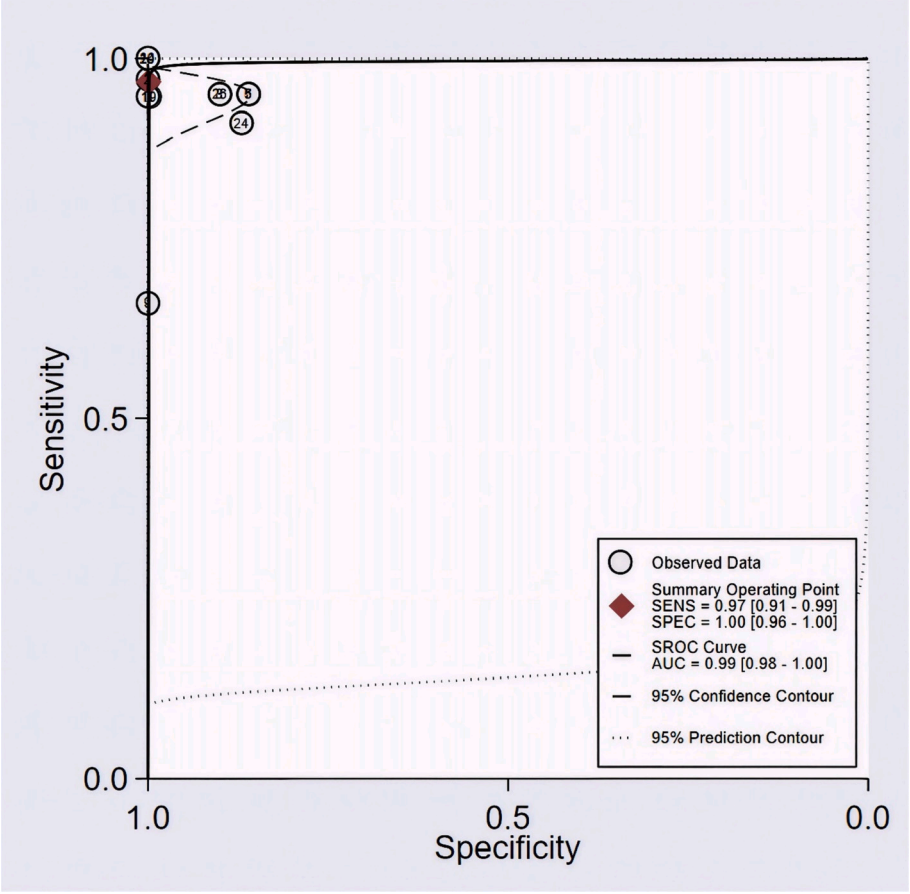


Fig. 4. Summary receiver operating characteristic curve showing the diagnostic performance of colonoscopy for colorectal cancer detection among asymptomatic average-risk adults aged 40–80 years in diagnostic accuracy studies identified from PubMed and Scopus, 4 November 2024 to 26 February 2025.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pmedr.2026.103551.

Data availability

The datasets generated and/or analyzed during the current study are not openly accessible but may be obtained from the corresponding author upon reasonable request.

References

Akimoto, N., Ugai, T., Zhong, R., Hamada, T., Fujiyoshi, K., Giannakis, M., et al., 2021. Rising incidence of early-onset colorectal cancer - a call to action. Nat. Rev. Clin. Oncol. 18 (4), 230–243. https://doi.org/10.1038/s41571-020-00445-1.

American Cancer Society. Colorectal cancer Screening Tests [Internet]. Atlanta: American Cancer Society; 2024 [cited Jul 03]. Available from: https://www.cdc.gov/colorectal-cancer/screening/index.html.

Areia, M., Fuccio, L., Hassan, C., Dekker, E., Dias-Pereira, A., Dinis-Ribeiro, M., 2019. Cost-utility analysis of colonoscopy or faecal immunochemical test for population-based organised colorectal cancer screening. United European Gastroenterol J 7 (1), 105–113. https://doi.org/10.1177/2050640618803196.

Aronsson, M., Carlsson, P., Levin, L.A., Hager, J., Hultcrantz, R., 2017. Cost-effectiveness of high-sensitivity faecal immunochemical test and colonoscopy screening for colorectal cancer. Br. J. Surg. 104 (8), 1078–1086. https://doi.org/10.1002/bjs.10536.

Barouni, M., Larizadeh, M.H., Sabermahani, A., Ghaderi, H., 2012. Markov's modeling for screening strategies for colorectal cancer. Asian Pac. J. Cancer Prev. 13 (10), 5125–5129. https://doi.org/10.7314/apjcp.2012.13.10.5125.

Canévet, M., Pruvost-Couvreur, M., Morvan, M., Badic, B., Brousse-Potocki, J., Kermarrec, T., et al., 2023. Sensitivity of fecal immunochemical test and risk factors for interval colorectal cancer in a French population. Clin. Res. Hepatol. Gastroenterol. 47 (3), 102093. https://doi.org/10.1016/j.clinre.2023.102093.

Cenin, D., Li, P., Wang, J., de Jonge, L., Yan, B., Tao, S., et al., 2022. Optimising colorectal cancer screening in Shanghai, China: a modelling study. BMJ Open 12 (5), e048156. https://doi.org/10.1136/bmjopen-2020-048156.

Cenin, D.R., Naber, S.K., de Weerdt, A.C., Jenkins, M.A., Preen, D.B., Ee, H.C., et al., 2020. Cost-effectiveness of personalized screening for colorectal Cancer based on polygenic risk and family history. Cancer Epidemiol. Biomarkers Prevent.: Publ. Am. Assoc. Cancer Res. Cospons. Am. Soc. Prevent. Oncol. 29 (1), 10–21. https://doi.org/10.1158/1055-9965.EPI-18-1123.

Dharwadkar, P., Zaki, T.A., Murphy, C.C., 2022. Colorectal Cancer in younger adults. Hematol. Oncol. Clin. North Am. 36 (3), 449–470. https://doi.org/10.1016/j.hoc.2022.02.005.

Fisher, D.A., Saoud, L., Hassmiller Lich, K., Fendrick, A.M., Ozbay, A.B., Borah, B.J., et al., 2021. Impact of screening and follow-up colonoscopy adenoma sensitivity on colorectal cancer screening outcomes in the CRC-AIM microsimulation model. Cancer Med. 10 (8), 2855–2864. https://doi.org/10.1002/cam4.3662.

Goede, S.L., Rabeneck, L., van Ballegooijen, M., Zauber, A.G., Paszat, L.F., Hoch, J.S., et al., 2017. Harms, benefits and costs of fecal immunochemical testing versus guaiac fecal occult blood testing for colorectal cancer screening. PLoS ONE 12 (3), e0172864. https://doi.org/10.1371/journal.pone.0172864.

Heitman, S.J., Hilsden, R.J., Au, F., Dowden, S., Manns, B.J., 2010. Colorectal cancer screening for average-risk north Americans: an economic evaluation. PLoS Med. 7 (11), e1000370. https://doi.org/10.1371/journal.pmed.1000370.

Hotta, K., Matsuda, T., Kakugawa, Y., Ikematsu, H., Kobayashi, N., Kushima, R., et al., 2016. Regional colorectal cancer screening program using colonoscopy on an island: a prospective Nii-jima study. Jpn. J. Clin. Oncol. 47 (2), 118–122. https://doi.org/10.1093/jjco/hyw155.

Jahn, B., Sroczynski, G., Bundo, M., Muhlberger, N., Puntscher, S., Todorovic, J., et al., 2019. Effectiveness, benefit harm and cost effectiveness of colorectal cancer screening in Austria. BMC Gastroenterol. 19 (1), 209. https://doi.org/10.1186/s12876-019-1121-y.

Joseph, D.A., King, J.B., Dowling, N.F., Thomas, C.C., Richardson, L.C., 2020. Vital signs: colorectal Cancer screening test use - United States, 2018. MMWR Morb. Mortal Wkly. Rep. 69 (10), 253–259. https://doi.org/10.15585/mmwr.mm6910a1.

Lee, J.K., Liles, E.G., Bent, S., Levin, T.R., Corley, D.A., 2014. Accuracy of fecal immunochemical tests for colorectal cancer: systematic review and meta-analysis. Ann. Intern. Med. 160 (3), 171. https://doi.org/10.7326/M13-1484.

Liberati, A., Altman, D.G., Tetzlaff, J., Mulrow, C., Gotzsche, P.C., Ioannidis, J.P., et al., 2009. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration. BMJ 339, b2700. https://doi.org/10.1136/bmj.b2700.

Lin, J.S., Perdue, L.A., Henrikson, N.B., Bean, S.I., Blasi, P.R., 2021. Screening for colorectal Cancer: updated evidence report and systematic review for the US preventive services task force. Jama 325 (19), 1978–1998. https://doi.org/10.1001/jama.2021.4417.

Lu, B., Luo, J., Yan, Y., Zhang, Y., Luo, C., Li, N., et al., 2024. Evaluation of long-term benefits and cost-effectiveness of nation-wide colorectal cancer screening strategies in China in 2020-2060: a modelling analysis. Lancet Reg. Health Western Pacific 51, 101172. https://doi.org/10.1016/j.lanwpc.2024.101172.

Maida, M., Dahiya, D.S., Shah, Y.R., Tiwari, A., Gopakumar, H., Vohra, I., et al., 2024. Screening and surveillance of colorectal cancer: a review of the literature. Cancers (Basel) 16, 2746. https://doi.org/10.3390/cancers16152746.

Melnitchouk, N., Soeteman, D.I., Davids, J.S., Fields, A., Cohen, J., Noubary, F., et al., 2018. Cost-effectiveness of colorectal cancer screening in Ukraine. Cost Effectiv. Resource Alloc.: C/E 16, 20. https://doi.org/10.1186/s12962-018-0104-0.

Nascimento de Lima, P., Matrajt, L., Coronado, G., Escaron, A.L., Rutter, C.M., 2025. Cost-effectiveness of noninvasive colorectal Cancer screening in community clinics. JAMA Netw. Open 8 (1), e2454938. https://doi.org/10.1001/jamanetworkopen.2024.54938.

Phisalprapa, P., Supakankunti, S., Chaiyakunapruk, N., 2019. Cost-effectiveness and budget impact analyses of colorectal cancer screenings in a low- and middle-income country: example from Thailand. J. Med. Econ. 22 (12), 1351–1361. https://doi.org/10.1080/13696998.2019.1674065.

Rex, D.K., Boland, C.R., Dominitz, J.A., Giardiello, F.M., Johnson, D.A., Kaltenbach, T., et al., 2017. Colorectal Cancer screening: recommendations for physicians and patients from the U.S. multi-society task force on colorectal Cancer. Am. J. Gastroenterol. 112 (7), 1016–1030. https://doi.org/10.1038/ajg.2017.174.

Sekiguchi, M., Igarashi, A., Matsuda, T., Matsumoto, M., Sakamoto, T., Nakajima, T., et al., 2016. Optimal use of colonoscopy and fecal immunochemical test for population-based colorectal cancer screening: a cost-effectiveness analysis using Japanese data. Jpn. J. Clin. Oncol. 46 (2), 116–125. https://doi.org/10.1093/jjco/hyv186.

Shaukat, A., Levin, T.R., 2022. Current and future colorectal cancer screening strategies. Nat. Rev. Gastroenterol. Hepatol. 19 (8), 521–531. https://doi.org/10.1038/s41575-022-00612-y.

Starker, A., Buttmann-Schweiger, N., Kraywinkel, K., Kuhnert, R., 2017. The utilization of colonoscopy in Germany. J. Health Monit. 2 (4), 76–81. https://doi.org/10.17886/rki-gbe-2017-126.

Sung, H., Ferlay, J., Siegel, R.L., Laversanne, M., Soerjomataram, I., Jemal, A., et al., 2021. Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J. Clin. 71 (3), 209–249. https://doi.org/10.3322/caac.21660.

Tran, T.N., Peeters, M., Hoeck, S., Van Hal, G., Janssens, S., De Schutter, H., 2022. Optimizing the colorectal cancer screening programme using faecal immunochemical test (FIT) in Flanders, Belgium from the “interval cancer” perspective. Br. J. Cancer 126 (7), 1091–1099. https://doi.org/10.1038/s41416-021-01694-2.

Tsapournas, G., Hellstrom, P.M., Cao, Y., Olsson, L.I., 2020. Diagnostic accuracy of a quantitative faecal immunochemical test vs. symptoms suspected for colorectal cancer in patients referred for colonoscopy. Scand. J. Gastroenterol. 55 (2), 184–192. https://doi.org/10.1080/00365521.2019.1708965.

Van den Puttelaar, R., Nascimento de Lima, P., Knudsen, A.B., Rutter, C.M., Kuntz, K.M., de Jonge, L., et al., 2024. Effectiveness and cost-effectiveness of colorectal Cancer screening with a blood test that meets the Centers for Medicare & Medicaid Services Coverage Decision. Gastroenterology 167 (2), 368–377. https://doi.org/10.1053/j.gastro.2024.02.012.

Van der Vlugt, M., Carvalho, B., Fliers, J., Montazeri, N., Rausch, C., Grobbee, E.J., et al., 2022. Missed colorectal cancers in a fecal immunochemical test-based screening program: molecular profiling of interval carcinomas. World J. Gastrointest. Oncol. 14 (11), 2195–2207. https://doi.org/10.4251/wjgo.v14.i11.2195.

Vuik, F.E., Nieuwenburg, S.A., Bardou, M., Lansdorp-Vogelaar, I., Dinis-Ribeiro, M., Bento, M.J., et al., 2019. Increasing incidence of colorectal cancer in young adults in Europe over the last 25 years. Gut 68 (10), 1820–1826. https://doi.org/10.1136/gutjnl-2018-317592.

Zauber, A.G., 2010. Cost-effectiveness of colonoscopy. Gastrointest. Endosc. Clin. N. Am. 20 (4), 751–770. https://doi.org/10.1016/j.giec.2010.07.008.